

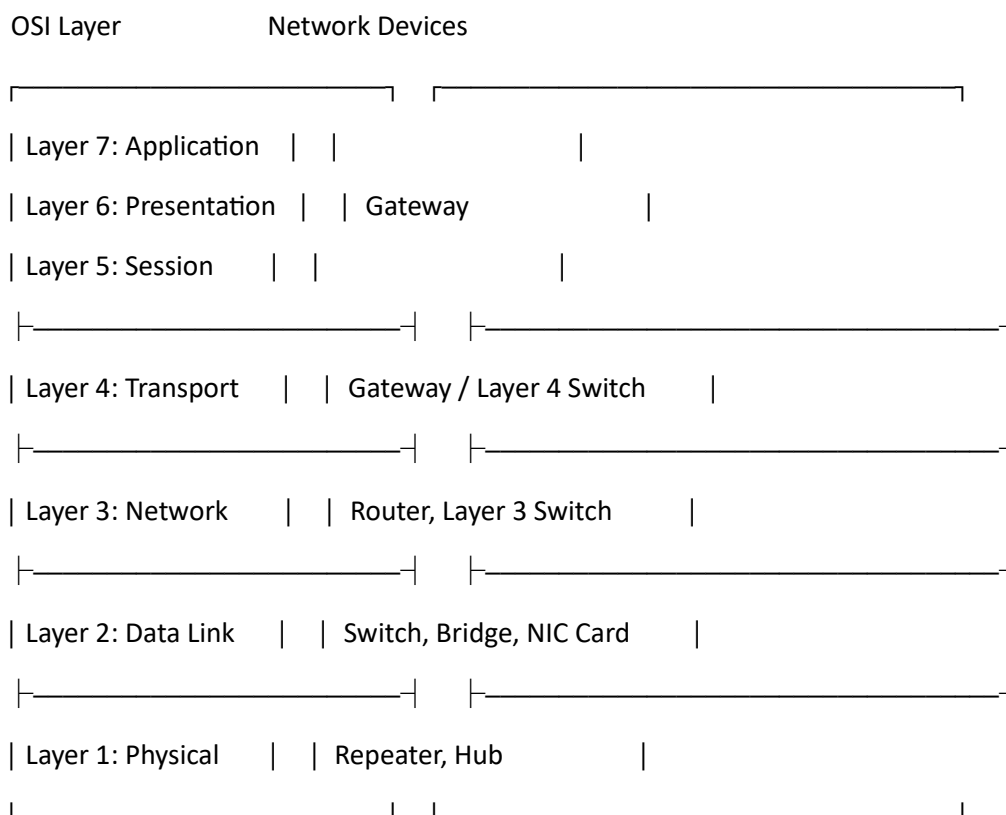
UNIT IV: NETWORKING DEVICES AND ADVANCED NETWORKING

Topics Covered

- **4.1** Network Devices: NIC Cards, Switches, Repeaters, Bridges, Gateways, Routers
- **4.2** Wireless Technologies: Wi-Fi vs. WiMAX
- **4.3** Cloud Computing Concepts, Service Models, and Deployment Models
- **4.4** Internet of Things (IoT) Architecture and Applications

4.1 NETWORK DEVICES

Networking devices (hardware components) connect computers, peripherals, and other networks together to enable data transfer and communication.



1. Network Interface Card (NIC)

- **OSI Layer:** Layer 1 (Physical) & Layer 2 (Data Link)
- **Function:** A hardware circuit board or chip installed in a computer that allows it to connect to a network (wired via Ethernet or wireless via Wi-Fi).
- **Key Concept:** Every NIC contains a unique, factory-burned 48-bit **MAC Address** (Media Access Control address) used for physical node-to-node identification.

2. Repeater

- **OSI Layer:** Layer 1 (Physical)

- **Function:** Receives weak or attenuated electrical, optical, or wireless signals, amplifies/regenerates them to original strength, and retransmits them over long cabling distances.
- **Key Concept:** Does not inspect frame headers or IP addresses; blindly amplifies all incoming bits (and noise).

3. Bridge

- **OSI Layer:** Layer 2 (Data Link)
- **Function:** Connects two separate LAN segments running the same protocol, filtering traffic based on destination MAC addresses.
- **Key Concept:** Maintains a MAC address table to segment collision domains and reduce network congestion.

4. Switch

- **OSI Layer:** Layer 2 (Data Link) *(Note: Layer 3 Switches also perform IP routing)*
- **Function:** A multi-port device that receives data frames and forwards them directly to the intended destination port using MAC address matching.
- **Key Concept:** Provides dedicated bandwidth per port, creates separate **collision domains** for each connected device, and operates in full-duplex mode.

5. Router

- **OSI Layer:** Layer 3 (Network)
- **Function:** Interconnects different networks (e.g., connecting a private LAN to the public Internet) and routes data packets along the best path using logical **IP addresses**.
- **Key Concept:** Uses routing tables and algorithms (e.g., OSPF, BGP) to forward traffic. Defines **broadcast domain boundaries** (blocks broadcast traffic).

6. Gateway

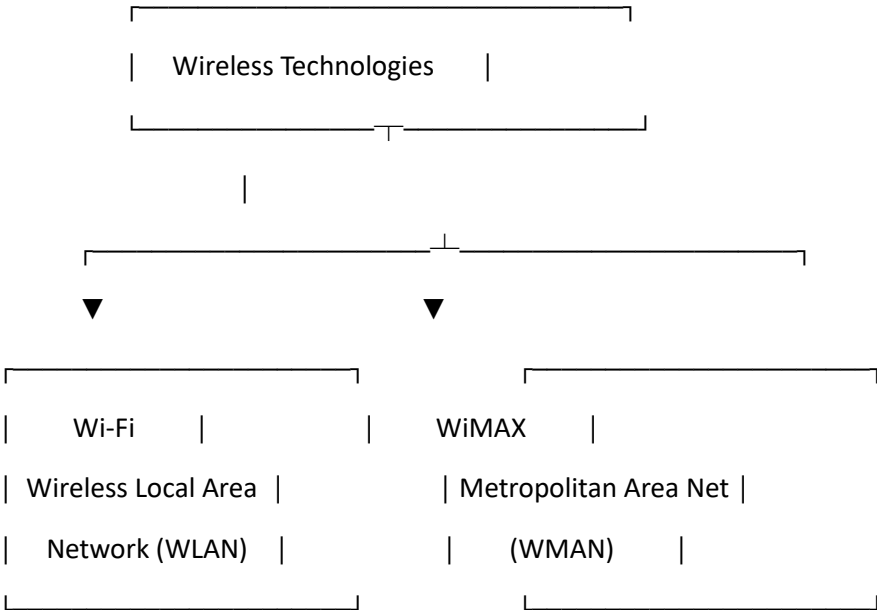
- **OSI Layer:** Operates across all layers (Layers 4 through 7)
- **Function:** Acts as a protocol converter or translator between two networks that use entirely different protocol suites (e.g., converting a TCP/IP network connection to an SNA or telecom network connection).
- **Key Concept:** Performs deep payload inspection and protocol translation at higher layers.

Comprehensive Device Comparison Matrix

Device	Primary OSI Layer	Addressing Used	Function / Role	Separates Collision Domains?	Separates Broadcast Domains?
Repeater	Layer 1	None (Bits)	Signal amplification	No	No
NIC Card	Layer 1 & 2	MAC Address	Physical interface to network	N/A	N/A
Bridge	Layer 2	MAC Address	Filters/connects 2 LAN segments	Yes	No
Switch	Layer 2	MAC Address	Multi-port frame switching	Yes (per port)	No
Router	Layer 3	IP Address	Packet routing across networks	Yes	Yes
Gateway	Layer 4 – 7	Multiple / Payload	Converts incompatible protocols	Yes	Yes

4.2 WI-FI VS. WIMAX

Both **Wi-Fi** and **WiMAX** are wireless networking technologies designed for high-speed data communications, but they differ significantly in range, coverage area, and intended use cases.



1. Wi-Fi (Wireless Fidelity)

- **Standard:** IEEE 802.11 family (e.g., 802.11a/b/g/n/ac/ax/be).
- **Scope:** Wireless Local Area Network (**WLAN**).
- **Purpose:** Provides short-range wireless internet connectivity for devices inside homes, offices, coffee shops, and campus buildings.

2. WiMAX (Worldwide Interoperability for Microwave Access)

- **Standard:** IEEE 802.16.
- **Scope:** Wireless Metropolitan Area Network (**WMAN**).
- **Purpose:** Provides long-range, broadband wireless internet access across large geographic areas (cities, rural regions) as an alternative to cable or DSL backhuls ("Last-Mile Broadband").

Detailed Comparison Table: Wi-Fi vs. WiMAX

Feature Parameter	Wi-Fi (IEEE 802.11)	WiMAX (IEEE 802.16)
Full Name	Wireless Fidelity	Worldwide Interoperability for Microwave Access
Network Type	Wireless LAN (WLAN)	Wireless MAN (WMAN)
Coverage Range	Short Range: $\approx$ 30 to 100 meters	Long Range: $\approx$ 10 to 50 kilometers
Data Transfer Speed	High local speeds (up to several Gbps with Wi-Fi 6/7)	Up to 100 Mbps (depending on distance/LOS)
Frequency Spectrum	Unlicensed bands (2.4 GHz, 5 GHz, 6 GHz)	Licensed and Unlicensed bands (2 GHz to 66 GHz)
Architecture	Point-to-Multipoint (Hub/AP to devices)	Point-to-Point & Point-to-Multipoint (Base Station to Tower)
Quality of Service (QoS)	Contention-based (CSMA/CA); basic QoS	Built-in hardware QoS guaranteed by time slots

Feature Parameter	Wi-Fi (IEEE 802.11)	WiMAX (IEEE 802.16)
Primary Usage	Home LANs, office wireless access, local hot spots	Last-mile broadband delivery, rural Internet, city mesh

4.3 CLOUD COMPUTING

What is Cloud Computing?

Cloud Computing is the on-demand availability and delivery of computing resources—including processing power, database storage, applications, and networking—over the Internet ("the cloud") on a pay-as-you-go pricing model.

Cloud Service Models (SPI Framework)

Cloud computing resources are categorized into three primary service delivery models:

SaaS (Software as a Service)	
[End Users] -> Gmail, Salesforce, Microsoft 365, Google Docs	
PaaS (Platform as a Service)	
[Developers] -> AWS Elastic Beanstalk, Heroku, Google App	
IaaS (Infrastructure as a Service)	
[SysAdmins] -> AWS EC2, Google Compute Engine, Azure VMs	

1. Infrastructure as a Service (IaaS):

- **Provides:** Virtualized basic computing infrastructure (raw Virtual Machines, servers, virtual networks, block storage).
- **User Responsibility:** Operating Systems, runtime environments, middleware, data, applications.
- **Examples:** Amazon EC2, Google Compute Engine (GCE), Microsoft Azure VMs.

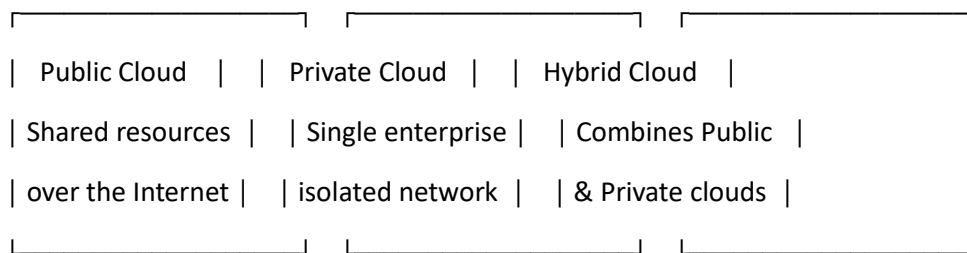
2. Platform as a Service (PaaS):

- **Provides:** A hardware and software platform tailored for application development, hosting, and testing (OS, database engines, language runtimes).
- **User Responsibility:** Application code and application configuration.
- **Examples:** AWS Elastic Beanstalk, Heroku, Google App Engine.

3. Software as a Service (SaaS):

- **Provides:** Fully managed end-user software applications accessible over a web browser or client interface.
- **User Responsibility:** Application configurations and user data management.
- **Examples:** Google Workspace (Gmail/Docs), Microsoft 365, Salesforce, Dropbox.

Cloud Deployment Models

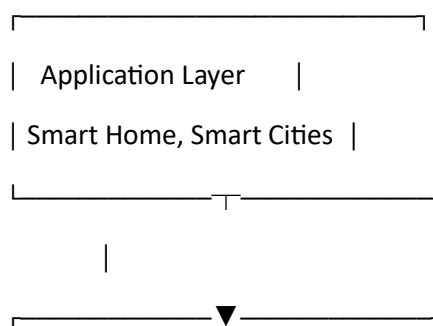


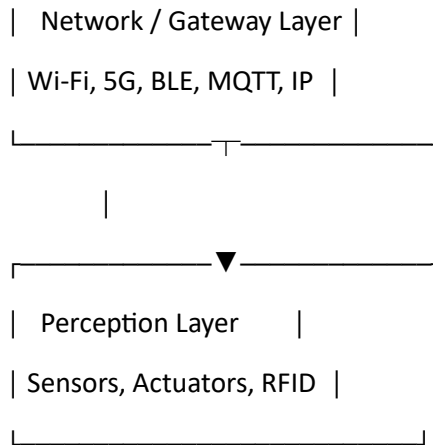
1. **Public Cloud:** Resources are owned and operated by third-party cloud service providers and delivered over the public Internet to multiple client organizations ("multitenant architecture").
 - *Examples:* AWS, Microsoft Azure, Google Cloud Platform (GCP).
2. **Private Cloud:** Cloud infrastructure dedicated exclusively to a single organization. It can be hosted on-premise in the company's datacenter or managed off-site by a third party.
 - *Advantages:* Maximum control, high security, strict compliance.
3. **Hybrid Cloud:** Combines public and private cloud environments using automated integration technologies, allowing data and applications to be shared between them (e.g., using private cloud for sensitive database storage and public cloud for auto-scaling web traffic).
4. **Community Cloud:** Infrastructure shared by several organizations with shared concerns (e.g., compliance requirements, security standards, industry regulation).

4.4 INTERNET OF THINGS (IOT)

What is IoT?

The **Internet of Things (IoT)** refers to a network of physical objects ("things") embedded with sensors, processing software, actuators, and communication modules that allow them to collect, exchange, and act on data over the Internet without human intervention.





Key Components of IoT Architecture

1. Perception / Sensing Layer (Edge Layer):

- **Sensors:** Collect raw data from the physical environment (e.g., temperature, humidity, motion, light, pressure sensors).
- **Actuators:** Execute physical actions based on control signals (e.g., opening a valve, turning on a motor, switching off lights).

2. Network / Processing Gateway Layer:

- Receives data from edge devices and routes it securely to edge computing nodes or central cloud systems.
- **Communication Protocols:** Wi-Fi, Bluetooth Low Energy (BLE), Zigbee, LoRaWAN, Cellular 4G/5G, **MQTT** (Message Queuing Telemetry Transport), **CoAP** (Constrained Application Protocol).

3. Application & Processing Layer:

- Analyzes gathered data using big data analytics, machine learning algorithms, and dashboard applications to drive automation or inform user decisions.

Real-World IoT Application Domains

- **Smart Homes:** Smart thermostats (Nest), automated lighting, connected smart locks, voice assistants.
- **Smart Cities:** Automated traffic light control, smart street lighting, waste management tracking, environmental pollution monitoring.
- **Industrial IoT (IIoT):** Predictive machine maintenance, assembly line automation, real-time supply chain tracking.
- **Healthcare:** Connected wearables for remote patient monitoring, automated insulin delivery systems, asset tracking in hospitals.

QUICK REVISION & SUMMARY

Key Technical Terms

- **Switch:** Layer 2 multi-port device that forwards frames using destination MAC addresses.
- **Router:** Layer 3 device that routes data packets across different networks using IP addresses.
- **Gateway:** Higher-layer device that converts protocols between incompatible systems.
- **Wi-Fi (IEEE 802.11):** Short-range wireless local network standard.
- **WiMAX (IEEE 802.16):** Long-range metropolitan wireless broadband standard.
- **IaaS / PaaS / SaaS:** The three primary cloud service delivery layers (Infrastructure, Platform, Software).
- **IoT (Internet of Things):** Network of physical devices embedded with sensors and actuators that collect and share data automatically.

IMPORTANT EXAM QUESTIONS

Short Answer Questions

1. Differentiate between a Layer 2 Switch and a Layer 3 Router based on OSI layers and addressing mechanisms.
2. What is a MAC address, where is it stored, and how many bits does it contain?
3. State two primary differences between Wi-Fi and WiMAX.
4. Define the 3 primary Cloud Computing service models (IaaS, PaaS, SaaS) with one example of each.
5. What is the role of sensors and actuators in an Internet of Things (IoT) environment?

Long Answer Questions

1. **Network Hardware Deep-Dive:** Describe the working principles, OSI layers, addressing schemes, and usage scenarios for Repeaters, Switches, Routers, and Gateways. Explain how collision domains and broadcast domains differ across these devices.
2. **Cloud Computing Framework:** Explain Cloud Computing in detail. Compare Public, Private, and Hybrid cloud deployment models across security, control, scalability, and cost.
3. **IoT Architecture & Applications:** Detail the multi-layer architecture of an Internet of Things (IoT) system. Describe the protocols used at the network layer and discuss real-world IoT use cases in smart cities and industrial automation.